

Multimodal Detection of Fraudulent Online Marketplace Listings Using Image and Text Analysis

David Ajayi
School of Computing
Dublin City University
Dublin, Ireland
david.ajayi3@mail.dcu.ie

Harmeek Singh Wazir
School of Computing
Dublin City University
Dublin, Ireland
harmeeksingh.wazir2@mail.dcu.ie

Abstract—The rapid advancement of generative artificial intelligence has significantly increased the realism of synthetic images and text, creating new challenges for online marketplace platforms in identifying fraudulent product listings. Existing approaches often focus on a single modality, such as image or text analysis, limiting their ability to detect increasingly sophisticated AI-generated content. This dissertation proposes a multimodal fraud-detection framework that integrates image authenticity, text authenticity and semantic consistency to improve the identification of synthetic marketplace listings.

The proposed framework combines NPR Deepfake and SAFE for AI-generated image detection, Binoculars for AI-generated text detection, and CLIP for evaluating semantic consistency between product images and descriptions. A custom marketplace dataset containing genuine and AI-generated listings was developed to evaluate the framework alongside the CNNDetection benchmark. Multiple decision-level fusion strategies, including logical voting and weighted fusion, were investigated to combine the outputs of the individual detectors.

Experimental results demonstrated that image authenticity provided the strongest indicator of synthetic marketplace fraud, with NPR Deepfake achieving the highest overall detection performance across the evaluated datasets. Although multimodal fusion did not consistently outperform the best individual detector, it successfully integrated complementary information from multiple modalities into a unified decision-making framework. The evaluation also showed that semantic consistency should be interpreted as supporting contextual evidence rather than a direct indicator of AI-generated content.

Overall, this research demonstrates the potential of multimodal approaches for marketplace fraud detection and provides a flexible foundation for future systems capable of addressing increasingly sophisticated AI-generated marketplace deception.

I. INTRODUCTION

Online marketplaces have become one of the primary ways people buy and sell products, offering access to millions of listings across a wide range of categories. As these platforms continue to grow, they also face an increasing number of fraudulent and misleading listings that can reduce consumer trust and result in financial loss. Sellers can deliberately upload incorrect product images, provide misleading descriptions, advertise counterfeit goods, or combine genuine content with AI-

generated material to make fraudulent listings appear authentic. The rapid development of generative artificial intelligence has further increased this challenge by making it possible to create highly realistic product images and convincing textual descriptions with minimal effort.

Detecting fraudulent marketplace listings is a challenging task because misleading information can appear in different forms. In some cases, the product image may not match the accompanying title or description. In others, the image itself may be AI-generated while the textual information appears genuine, or vice versa. These inconsistencies are often difficult to identify using methods that analyse only images or only text. Consequently, relying on a single source of information may not provide sufficient evidence to distinguish genuine listings from fraudulent ones.

Recent advances in multimodal learning have created new opportunities for analysing images and text together. Vision-language models are capable of learning relationships between visual and textual information, while specialised detectors have been developed to identify AI-generated images and machine-generated text. Although these techniques have demonstrated strong performance individually, there is still limited understanding of how they can be effectively combined for marketplace fraud detection.

The aim of this research is to investigate whether combining image authenticity, text authenticity and semantic consistency within a multimodal framework can improve the detection of fraudulent and misleading marketplace listings. To achieve this, the research evaluates AI-generated image detection methods, AI-generated text detection methods, semantic image-text consistency analysis, and multiple decision-level fusion strategies to determine how complementary sources of evidence can be integrated into a unified fraud-detection framework.

The outcome of this research is a multimodal fraud detection framework that combines image authenticity using NPR Deepfake [16] and SAFE [16], text authenticity using Binoculars [5], and semantic image-text consistency using CLIP [1] to identify suspicious marketplace listings. The

proposed framework is evaluated using both publicly available datasets and a custom marketplace dataset developed specifically for this research, providing a comprehensive assessment of individual detection methods and multimodal fusion strategies for identifying fraudulent online marketplace listings.

II. RELATED WORK

A. Fraud Detection in Online Marketplaces

Online marketplaces are increasingly affected by fraudulent and misleading listings as the volume of user-generated content grows. Earlier fraud detection systems relied mainly on rules and manually engineered features, while recent approaches use machine learning to analyse textual, visual and behavioural information. Nguyen *et al.* [19] proposed a multimodal framework for detecting fraudulent real-estate advertisements by jointly analysing images and text. Zhang *et al.* [21] introduced AIGuard, a benchmark for evaluating AI-generated content risks in e-commerce. Other studies have examined graph-based relationships between users and transactions [26] and multimodal fake-review detection using textual and visual features [10], [22]. These studies indicate that marketplace fraud can involve multiple forms of evidence and may benefit from multimodal analysis.

B. AI-Generated Image Detection

Advances in generative image models have made synthetic images increasingly difficult to distinguish from genuine photographs. Wang *et al.* [18] demonstrated that convolutional neural networks could detect GAN-generated images by identifying subtle generation artefacts. Later approaches focused on improving generalisation across unseen generators. DIRE [17] uses reconstruction error to detect diffusion-generated images, while PatchCraft [15] analyses local texture patches. Li *et al.* [16] proposed image transformations to improve cross-generator robustness, and Gye *et al.* [23] addressed content bias in synthetic-image detection. Other work has explored multimodal foundation models for explainable fake-image detection [8] and CLIP-based universal detection through GCS-Net [27]. However, detector performance can vary across datasets and image domains, motivating evaluation on marketplace-specific content.

C. AI-Generated Text Detection

The widespread use of large language models has increased interest in identifying machine-generated text. DetectGPT [4] introduced a zero-shot method based on probability curvature, while Fast-DetectGPT [13] improved efficiency through conditional probability estimation. Binoculars [5] detects generated text by comparing the behaviour of paired language models, whereas RADAR [14] uses adversarial learning to improve robustness against rewritten content. Tufts *et al.* [6] showed that detector performance varies across language models, domains, writing styles and text lengths. This is particularly relevant to marketplaces, where short and specification-based descriptions may resemble AI-generated text. Text authenticity should therefore be treated as supporting evidence rather than a definitive indicator of fraud.

D. Vision–Language Models and Image–Text Consistency

Vision–language models represent images and text within a shared semantic space. CLIP [1] uses contrastive learning to align visual and textual representations, allowing similarity to be measured between an image and its description. BLIP-2 [2] and LLaVA [3] extended multimodal capabilities by connecting visual encoders with large language models. Within e-commerce, CLIP has also been adapted for category-to-image retrieval [20]. Related approaches have applied semantic alignment to fake-news detection [9], [11], [24]. Although image–text consistency can indicate whether a description corresponds to an image, it cannot determine whether either component is genuine.

E. Multimodal Detection and Fusion

Multimodal detection combines evidence from different sources through feature-level or decision-level fusion. Decision-level fusion is useful when individual models perform complementary tasks, such as image authenticity detection, text authenticity detection and semantic consistency analysis. Studies involving fake advertisements, reviews and news have shown the value of combining visual and textual evidence [11], [19], [22], [24]. However, limited research has examined how specialised AI-image detectors, AI-text detectors and vision–language models can be combined within a unified marketplace fraud-detection framework.

F. Research Gap

AI-generated image detection, AI-generated text detection and image–text consistency are commonly investigated as separate problems. Existing marketplace research also tends to focus on individual modalities or general multimodal classification, with limited attention given to combining authenticity and semantic-consistency signals. Furthermore, few studies evaluate realistic marketplace scenarios in which AI-generated product images are paired with either human-written or AI-generated descriptions. This dissertation addresses these gaps by evaluating a decision-level multimodal framework that integrates image authenticity, text authenticity and semantic consistency for detecting synthetic marketplace listings.

III. METHODOLOGY

This research adopts a staged experimental methodology to investigate the detection of fraudulent and misleading online marketplace listings using multimodal artificial intelligence techniques. Five experimental stages were designed to evaluate semantic image–text consistency, multimodal fraud detection, CLIP fine-tuning, AI-generated content detection and multimodal decision-level fusion. Each stage investigates a different aspect of marketplace fraud detection and collectively provides the evidence required to answer the research questions.

A. Research Questions

The research was guided by the following questions:

- **RQ1:** Can semantic similarity between product images and textual information be used to distinguish genuine

marketplace listings from manipulated or mismatched listings?

- **RQ2:** Can multimodal representations support supervised and unsupervised approaches for fraud detection?
- **RQ3:** Does fine-tuning CLIP improve image–text consistency detection?
- **RQ4:** How accurately can existing AI-generated image and AI-generated text detection methods identify synthetic marketplace content?
- **RQ5:** Can combining semantic consistency, image authenticity and text authenticity provide a more reliable indicator of marketplace fraud than any individual detection approach?

B. Semantic Consistency Investigation

The first stage investigated whether semantic consistency between marketplace product images and their associated textual information could be used as an indicator of fraud. Genuine listings are expected to exhibit strong agreement between their visual and textual content, whereas manipulated listings often contain mismatched titles or descriptions. This stage established a semantic similarity baseline for distinguishing genuine listings from fraudulent or misleading listings.

C. Multimodal Fraud Detection

The second stage investigated whether multimodal feature representations could support fraud detection using supervised and unsupervised machine learning approaches. By learning patterns from genuine and manipulated marketplace listings, this stage evaluated the ability of multimodal representations to distinguish normal listings from potentially fraudulent ones.

D. Fine-Tuning Investigation

The third stage investigated whether adapting the CLIP model to a domain-specific dataset could improve image–text consistency detection. A fine-tuning experiment using visually similar smartphone products was conducted to evaluate whether domain adaptation enhances the model’s ability to distinguish subtle semantic inconsistencies between related products.

E. AI-Generated Content Detection

The fourth stage evaluated the effectiveness of existing AI-generated image and AI-generated text detection methods on marketplace content. Image detectors were evaluated using genuine and AI-generated product images, while text detectors were assessed using authentic and AI-generated marketplace descriptions. The objective was to evaluate the effectiveness and robustness of these methods when applied independently to realistic marketplace data.

F. Multimodal Fraud Detection Framework

The final stage evaluated whether combining semantic image–text consistency (CLIP), AI-generated image detection (NPR Deepfake and SAFE), and AI-generated text detection (Binoculars) within a decision-level fusion framework provides a more reliable indicator of marketplace fraud than

any individual detection method. The integrated framework combines complementary evidence from each component to produce an overall fraud assessment and is evaluated against the individual detectors to determine the effectiveness of multimodal fusion.

IV. EXPERIMENTAL SETUP

A. Experimental Overview

This section describes the datasets, models and evaluation procedures used to investigate the five research questions presented in the methodology. The experiments examined semantic image–text consistency, multimodal fraud detection using learned representations, domain-specific CLIP fine-tuning, AI-generated image and text detection, and the integration of these components within a unified multimodal fraud detection framework. Both publicly available datasets and custom-built marketplace datasets were used throughout the study.

B. Datasets

Several datasets were used to evaluate different aspects of fraudulent and misleading marketplace listings. The Amazon Berkeley Objects dataset provided genuine marketplace listings for the semantic consistency and multimodal fraud detection experiments. Additional datasets were created for manipulated listings, visually similar smartphone products, AI-generated product images and AI-generated marketplace descriptions.

1) *Amazon Berkeley Objects Dataset:* The Amazon Berkeley Objects dataset served as the primary source of genuine marketplace listings. It contains product images together with structured metadata, including titles, descriptions, categories and sub-categories, making it suitable for multimodal analysis.

The dataset was preprocessed to associate each valid product image with its corresponding textual information. Listings containing missing metadata, unavailable images or incomplete product information were removed. The cleaned data were then sampled to produce subsets containing approximately 10,000, 25,000, 50,000 and 100,000 listings. These subsets were used in the semantic consistency and multimodal fraud detection experiments.

2) *Marketplace Fraud Dataset:* A custom marketplace fraud dataset was developed because no suitable public dataset containing realistic image–text inconsistencies was available. Genuine listings from the cleaned Amazon Berkeley Objects dataset were manipulated by replacing titles, descriptions or both with information taken from different products.

This produced image–text pairs in which the visual and textual information referred to different products while maintaining realistic marketplace content. The resulting dataset contained both genuine and manipulated listings and was used for the semantic consistency and multimodal fraud detection experiments.

3) *Smartphone Fine-Tuning Dataset:* A separate smartphone dataset was created for the CLIP fine-tuning experiment. The dataset contained images and descriptions of visually

similar smartphone models, including both matched and deliberately mismatched image–text pairs.

Smartphones were selected because products within the same category often exhibit similar visual characteristics while differing in identity and specifications. This provided a challenging domain-specific dataset for evaluating whether fine-tuning improves the ability of CLIP to distinguish subtle semantic inconsistencies.

4) *AI-Generated Image Dataset*: A dedicated dataset was constructed to evaluate AI-generated image detection methods. Genuine product images were collected from online marketplaces, including Amazon, eBay and Facebook Marketplace. AI-generated versions of the selected products were then created using ChatGPT.

The dataset contained genuine and AI-generated images across multiple product categories and was used exclusively to evaluate image authenticity detectors.

5) *AI-Generated Text Dataset*: A separate dataset was developed to evaluate AI-generated text detection methods. Human-written product descriptions were collected from marketplace listings, while corresponding AI-generated descriptions were produced using ChatGPT.

The generated descriptions were designed to resemble realistic marketplace content in style, structure and level of product detail. The dataset contained both human-written and AI-generated descriptions across multiple product categories and was used exclusively to evaluate text authenticity detectors.

C. Experimental Models

The experimental models were organised according to the five research questions, with each experiment investigating a different aspect of fraudulent marketplace listing detection.

1) *RQ1: Semantic Consistency Investigation*: To investigate whether semantic image–text consistency could distinguish genuine listings from manipulated listings, the CLIP model with the ViT-B/32 backbone was employed.

Image embeddings were extracted from each product image, while text embeddings were generated from the corresponding title and sub-category. Cosine similarity was computed between the image embedding and each textual representation.

The final semantic consistency score was calculated using a weighted combination of 70% title similarity and 30% sub-category similarity. Greater weight was assigned to the title because it provides the primary description of the product, whereas the sub-category offers complementary contextual information. Higher scores indicated stronger agreement between the visual and textual information, while lower scores suggested potential inconsistencies.

2) *RQ2: Fraud Detection Using Multimodal Representations*: RQ2 investigated whether multimodal feature representations extracted by CLIP could support supervised and unsupervised fraud detection.

For the supervised experiment, a balanced dataset consisting of 1,000 genuine and 1,000 manipulated listings was created. Image and text embeddings were extracted using CLIP and

concatenated to form multimodal feature vectors, which were used to train a Random Forest classifier.

For the unsupervised experiment, One-Class Support Vector Machine and Isolation Forest models were trained primarily using genuine marketplace listings. These models learned the distribution of normal marketplace representations and assigned anomaly scores to unseen listings. Genuine and manipulated listings were then compared to evaluate whether fraudulent samples were consistently identified as anomalous.

3) *RQ3: CLIP Fine-Tuning*: RQ3 investigated whether domain-specific fine-tuning improved CLIP’s ability to identify semantic inconsistencies between visually similar products.

The pre-trained CLIP ViT-B/32 model was fine-tuned using the smartphone dataset containing matched image–text pairs. Following fine-tuning, both the original and adapted models were evaluated using matched and mismatched smartphone pairs.

Cosine similarity scores were computed for both models, and identical decision thresholds were applied to classify image–text pairs. Performance comparisons determined whether domain adaptation improved semantic consistency detection.

4) *RQ4: AI-Generated Content Detection*: RQ4 evaluated existing AI-generated image and AI-generated text detection methods independently.

For AI-generated image detection, NPR Deepfake Detection [16], SAFE [16], PatchCraft [15] and FreqNet [16] were evaluated using genuine marketplace images and their corresponding AI-generated versions. The same image detectors were additionally evaluated using the CNNDetection benchmark to assess their ability to generalise beyond marketplace-specific images.

For AI-generated text detection, Binoculars [5], RoBERTa-Base, RoBERTa-Large and DetectGPT [4] were evaluated using human-written and AI-generated marketplace descriptions. Each detector produced predictions indicating whether the text was classified as human-written or AI-generated.

The objective of this stage was to compare the effectiveness of existing authenticity detectors and identify the strongest image and text detection methods for the final multimodal framework.

5) *RQ5: Multimodal Fraud Detection Framework*: RQ5 investigated whether combining semantic consistency, image authenticity and text authenticity provides a more reliable indicator of marketplace fraud than any individual detection approach.

The framework integrated four complementary components: CLIP semantic consistency, NPR Deepfake Detection, SAFE and Binoculars. Their outputs were combined using multiple decision-level fusion strategies, including logical voting and weighted fusion, to produce an overall fraud prediction.

The multimodal framework was then compared with each individual detector to determine whether combining complementary evidence improved the reliability of fraud detection.

D. Evaluation Metrics

Model performance was primarily evaluated using accuracy, precision, recall and F1-score. Accuracy measured the overall proportion of correctly classified samples, precision measured the proportion of predicted fraudulent or AI-generated samples that were correctly identified, recall measured the proportion of actual fraudulent or synthetic samples successfully detected, and the F1-score provided a balanced measure of precision and recall.

Confusion-matrix statistics, including true positives, true negatives, false positives and false negatives, were also reported where appropriate. For the CLIP-based experiments, cosine similarity distributions and decision thresholds were analysed to evaluate semantic consistency, while anomaly scores were examined for the unsupervised fraud detection experiments. Together, these evaluation measures provided a consistent basis for comparing the performance of the different approaches investigated throughout the dissertation.

V. RESULTS AND DISCUSSION

This section presents the experimental results obtained from the proposed multimodal fraud detection framework and discusses their implications for detecting fraudulent online marketplace listings. The results are organised according to the six research questions introduced in section 3. Each section presents the performance of the corresponding experiment, followed by an analysis of the findings and their relevance to marketplace fraud detection. The experiments investigate semantic consistency analysis, supervised and unsupervised learning, representation adaptation through fine-tuning, alternative feature representations, AI-generated content detection, and multimodal fusion.

A. RQ1: Semantic Consistency Analysis

The first research question investigated whether semantic similarity between product images and textual information could distinguish genuine marketplace listings from fraudulent or misleading listings. To evaluate this, 1,000 genuine product listings were selected from the Amazon Berkeley Objects dataset. These listings formed the genuine class, while a corresponding set of 1,000 fraudulent listings was created by retaining the product images but replacing their titles and subcategories with information from different products. This produced a balanced evaluation dataset containing 2,000 listings: 1,000 genuine and 1,000 synthetically mismatched listings.

CLIP ViT-B/32 was then used to encode each product image, title and sub-category. Cosine similarity was calculated between the image and each textual component. The final semantic similarity score was computed as a weighted combination of 70% title similarity and 30% sub-category similarity. These weights were selected empirically to give greater importance to the product title, which provides the primary description of the item, while the sub-category contributes complementary contextual information. Figure 1 presents the resulting score distributions. Genuine listings achieved a higher average

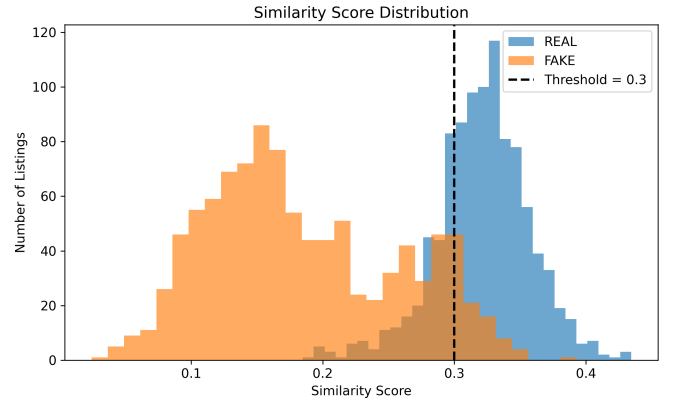


Fig. 1. Distribution of semantic similarity scores for genuine and mismatched marketplace listings.

TABLE I
PERFORMANCE OF THE SEMANTIC SIMILARITY APPROACH.

Metric	Value
Accuracy	84.45%
Precision	79.52%
Recall	92.80%
F1-score	85.65%
ROC-AUC	94.80%

similarity score of 0.321, compared with 0.184 for mismatched listings. Using a decision threshold of 0.30, listings below the threshold were classified as fraudulent and those at or above it were classified as genuine.

Table I summarises the classification performance obtained by comparing the predicted classes with the known genuine and fraudulent labels. The approach achieved an accuracy of 84.45%, precision of 79.52%, recall of 92.80%, F1-score of 85.65%, and ROC-AUC of 94.80%. The high recall shows that most of the synthetically mismatched listings were successfully detected, while the ROC-AUC indicates strong separation between the two classes across different thresholds.

Figure 2 presents the confusion matrix. Of the 1,000 genuine listings, 761 were correctly classified and 239 were incorrectly flagged as fraudulent. Of the 1,000 mismatched listings, 928 were correctly detected and 72 were incorrectly classified as genuine. This shows that the selected threshold favoured detecting fraudulent listings, although it also produced some false alerts for genuine listings.

Overall, the results provide a positive answer to RQ1. The lower similarity scores produced by the mismatched listings, together with the achieved classification performance, demonstrate that CLIP-based image–text consistency is an effective feature for identifying misleading online marketplace listings.

B. RQ2: Can Representation Learning Support the Detection of Anomalous Marketplace Listings?

This research question investigated whether multimodal representations learned exclusively from genuine marketplace listings could be used to identify anomalous or potentially

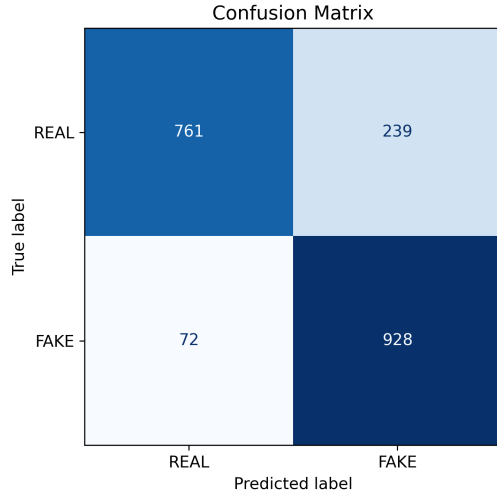


Fig. 2. Confusion matrix for semantic similarity-based fraud detection.

TABLE II
AVERAGE ONE-CLASS SVM NORMALITY SCORES ACROSS DIFFERENT AMAZON TRAINING SIZES.

Training Size	Genuine	Fraudulent	Difference
1,000	7.245	6.899	0.346
10,000	73.645	70.319	3.326
25,000	184.641	176.192	8.449
50,000	369.593	352.767	16.826
100,000	740.124	706.524	33.600

fraudulent listings. CLIP ViT-B/32 was used to extract 512-dimensional image embeddings and 512-dimensional text embeddings from genuine Amazon Berkeley Objects (ABO) listings. The two embeddings were concatenated to form a 1,024-dimensional multimodal representation for each listing. A One-Class Support Vector Machine (One-Class SVM) was trained using only genuine Amazon listings, while an Isolation Forest was evaluated for comparison. Both models were tested on 55 manually labelled listings from the custom evaluation dataset, comprising 32 genuine and 23 fraudulent listings.

Table II summarises the average One-Class SVM normality scores obtained using five different Amazon training-set sizes.

Across all five experiments, genuine listings consistently achieved higher normality scores than fraudulent listings. Because the raw One-Class SVM score scale increased as the training set became larger, the results were normalised relative to the genuine class mean to enable comparison across experiments.

Figure 3 shows that fraudulent listings consistently received normality scores that were approximately 95% of those assigned to genuine listings across all five training sizes. In other words, the average normality score for fraudulent listings was only about 5% lower than that of genuine listings. This indicates that the learned multimodal representation consistently regarded genuine listings as more normal, but the relatively small difference between the two classes changed



Fig. 3. Normalised mean One-Class SVM scores for genuine and fraudulent listings across different Amazon training sizes.

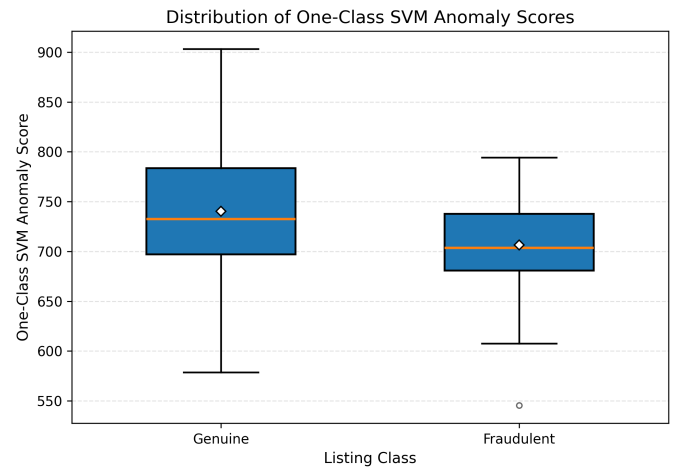


Fig. 4. Distribution of One-Class SVM normality scores for genuine and fraudulent listings using the 100,000-listing training set.

little as the training data increased. Consequently, increasing the number of training samples did not substantially improve the separation between genuine and fraudulent listings.

The distribution of scores obtained from the 100,000-listing experiment is shown in Fig. 4.

The boxplot shows that genuine listings achieved both higher median and higher mean normality scores than fraudulent listings. However, noticeable overlap exists between the two distributions, indicating that some fraudulent listings remained semantically similar to genuine products and therefore received relatively high normality scores. This suggests that representation learning alone provides a useful anomaly signal but cannot completely separate genuine and fraudulent marketplace listings.

To determine whether this behaviour was specific to the One-Class SVM, the same feature representations were evaluated using Isolation Forest. Figure 5 compares the relative separation achieved by both anomaly detection models.

The One-Class SVM achieved a relative separation of 4.54%, compared with 1.21% for Isolation Forest, indicat-

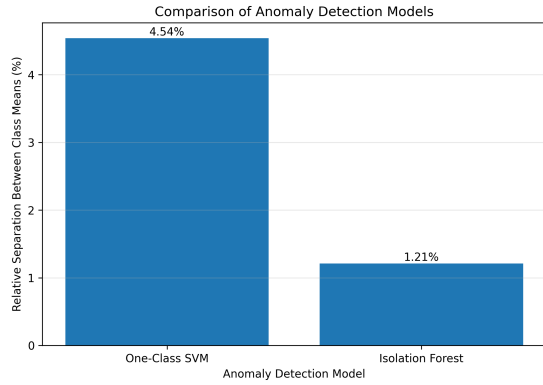


Fig. 5. Relative separation achieved by One-Class SVM and Isolation Forest.

TABLE III
PERFORMANCE COMPARISON BETWEEN THE ORIGINAL AND FINE-TUNED CLIP MODELS.

Model	Threshold	Accuracy	Precision	Recall	F1
Original CLIP	0.289	0.82	0.75	0.15	0.25
Fine-Tuned CLIP	0.014	0.20	0.20	1.00	0.33

ing that it was better suited to the CLIP-based multimodal representations used in this study. Although the One-Class SVM consistently assigned higher normality scores to genuine listings, the overlap observed in Fig. 4 demonstrates that anomaly detection alone is insufficient for reliably distinguishing all fraudulent listings. These findings suggest that representation learning provides a valuable anomaly signal but is most effective when combined with additional multimodal features and supervised classification methods.

C. RQ3: Does Fine-Tuning CLIP Improve Image–Text Consistency Detection?

This research question investigated whether fine-tuning CLIP on a domain-specific dataset could improve its ability to distinguish between visually similar marketplace products. A proof-of-concept experiment was conducted using five smartphone models from Apple and Samsung. Twenty smartphone images captured from multiple viewpoints were paired with their correct and incorrect model names, producing 20 matched and 80 mismatched image–text pairs. The larger number of mismatched pairs was intentionally created by pairing each image with multiple incorrect model names, allowing the experiment to evaluate the model’s ability to reject visually similar but semantically incorrect matches. Although this resulted in an imbalanced dataset, it better reflects the practical task of distinguishing a single correct match from several plausible incorrect alternatives. The original and fine-tuned CLIP models were evaluated on the same dataset using optimised similarity thresholds.

Table III and Figure 6 summarise the performance of both models. The original CLIP model achieved higher accuracy and precision, whereas the fine-tuned model achieved perfect recall. However, the higher recall was accompanied by substantially lower precision and accuracy, indicating that

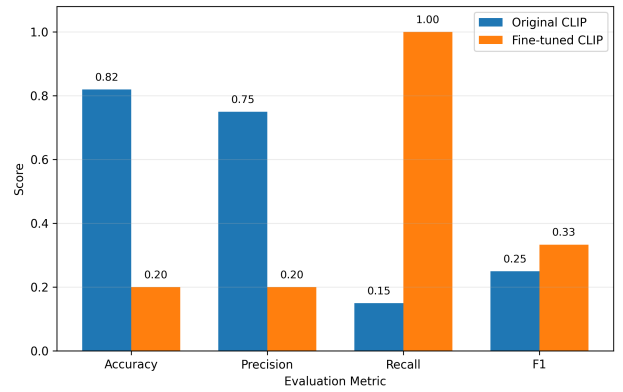


Fig. 6. Performance comparison between the original and fine-tuned CLIP models.

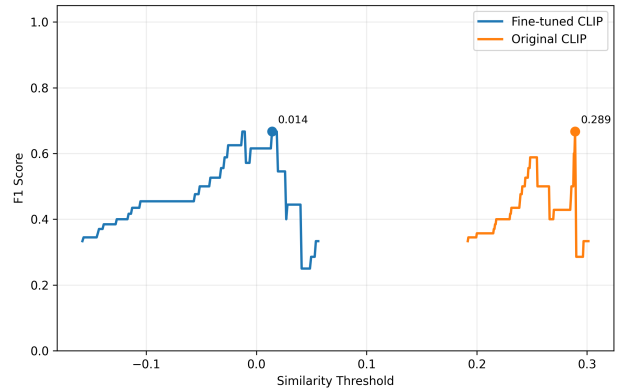


Fig. 7. Threshold optimisation for the original and fine-tuned CLIP models.

many mismatched pairs were incorrectly classified as matches. Although the F1 score increased from 0.25 to 0.33, the overall improvement was limited.

Threshold optimisation was performed to identify the similarity value that maximised the F1 score for each model. As shown in Figure 7, the fine-tuned model required a substantially lower threshold than the original model, demonstrating that fine-tuning altered the similarity-score distribution. Despite this optimisation, considerable overlap remained between matched and mismatched pairs, limiting reliable discrimination between visually similar smartphone models.

Overall, fine-tuning changed the behaviour of the CLIP model rather than producing a substantial performance improvement. The model became more sensitive to matched pairs but also generated more false positives. This result is likely influenced by the small dataset and the strong visual similarity between the selected smartphone models. Therefore, fine-tuning was treated as an exploratory investigation rather than the primary multimodal fraud-detection approach used in this research.

D. RQ4: How Accurately Can Existing AI-Generated Image and AI-Generated Text Detection Methods Identify Synthetic Marketplace Content?

This research question evaluated the effectiveness of existing AI-generated image and AI-generated text detection methods for identifying synthetic marketplace content. The evaluation was conducted in three stages. First, AI-image detectors were evaluated using a custom marketplace dataset and the CNNDetection benchmark. Second, AI-text detectors were evaluated using genuine and AI-generated marketplace descriptions. Finally, the strongest image and text detectors were combined to investigate whether multimodal fusion improved fraudulent marketplace listing detection.

1) *AI-Generated Image Detection:* Four AI-image detectors were evaluated on the expanded custom marketplace dataset containing 100 genuine and 100 AI-generated product images. The same detectors were subsequently evaluated on the CNNDetection benchmark containing 90,329 images to assess their generalisation performance.

Tables IV and V show that NPR Deepfake achieved the highest performance on the custom marketplace dataset, with an accuracy of 99.50% and a recall of 100.00%. SAFE achieved the strongest overall performance on the CNNDetection benchmark, obtaining the highest accuracy and F1-score. These results indicate that detector performance varied across datasets, with NPR Deepfake performing best on marketplace images and SAFE demonstrating stronger cross-domain generalisation.

2) *AI-Generated Text Detection:* Four AI-text detectors were evaluated using a balanced dataset containing 100 genuine marketplace descriptions and 100 AI-generated marketplace descriptions.

Table VI shows that Binoculars achieved the strongest overall performance, with an accuracy of 65.00%, a recall of 88.00%, and an F1-score of 71.54%. FastDetectGPT achieved the second-highest accuracy, while both RoBERTa models produced lower performance. Although Binoculars identified most AI-generated descriptions, it also produced a high number of false positives, showing that genuine marketplace descriptions were frequently classified as AI-generated.

3) *Multimodal Evaluation:* The strongest image detectors, NPR Deepfake and SAFE, were combined with Binoculars to investigate whether multimodal fusion improved marketplace fraud detection. Two experiments were conducted. Experiment 1 compared genuine listings against AI-generated images paired with convincing human-written descriptions. Experiment 2 compared genuine listings against AI-generated images paired with AI-generated descriptions.

Tables VII and VIII show that NPR Deepfake remained the strongest method in both experiments, achieving an accuracy of 99.50%. The multimodal majority-vote approach achieved accuracies of 98.00% and 98.50% in Experiments 1 and 2, respectively, but did not outperform NPR alone. Binoculars achieved higher recall when AI-generated descriptions were used, but its low specificity introduced additional false-positive predictions.

Overall, existing AI-image detectors substantially outperformed the evaluated AI-text detectors for identifying synthetic marketplace content. NPR Deepfake achieved the highest performance on the custom marketplace dataset, while SAFE demonstrated stronger generalisation on the larger CNNDetection benchmark. Although multimodal fusion produced high performance, it did not surpass the strongest image-only detector. This indicates that image authenticity provided the dominant signal for detecting synthetic marketplace listings, while text authenticity offered limited additional benefit.

VI. CONCLUSION AND FUTURE WORK

This dissertation proposed and evaluated a multimodal framework for detecting synthetic marketplace fraud by combining image authenticity, text authenticity and semantic consistency. The experimental results demonstrated that image-based detection methods provided the strongest evidence for identifying fraudulent marketplace listings, with NPR Deepfake achieving the highest overall performance across the evaluated datasets. While multimodal fusion successfully integrated complementary information from image, text and semantic analysis, it did not consistently outperform the best individual detector because NPR already achieved near-perfect accuracy. Nevertheless, the framework demonstrated that combining multiple modalities provides a structured and flexible approach to marketplace fraud detection and offers a foundation for future multimodal moderation systems.

Although the proposed framework produced promising results, several limitations should be acknowledged. The evaluation was conducted using a relatively small custom marketplace dataset and focused primarily on AI-generated fraud scenarios. In addition, only a limited number of image and text detection models were evaluated, while the semantic consistency experiments were performed using listings in which the descriptions intentionally matched the associated images.

Future work should evaluate the proposed framework using larger and more diverse real-world marketplace datasets containing a wider range of fraudulent listing types. Additional image and text detection models could also be incorporated, together with more advanced fusion techniques that dynamically adapt the contribution of each modality. Furthermore, integrating behavioural information, such as seller history, pricing anomalies and marketplace metadata, could further improve the robustness and practical applicability of multimodal fraud detection systems.

GENAI USAGE DISCLOSURE

Generative Artificial Intelligence (GenAI) tools were used during the development and preparation of this dissertation. ChatGPT by OpenAI was used to assist with language refinement, grammar correction, academic structuring, explanation of technical concepts, Python code development and debugging, interpretation of experimental results, and improvement of the presentation of tables and discussions.

GenAI assistance was applied in the Literature Review, Methodology, Implementation, Experimental Evaluation, Dis-

TABLE IV
AI-IMAGE DETECTOR PERFORMANCE ON THE EXPANDED CUSTOM MARKETPLACE DATASET.

Detector	Accuracy	Precision	Recall	F1-score	TN	FP	FN	TP
NPR Deepfake	99.50	99.01	100.00	99.50	99	1	0	100
SAFE	97.00	97.96	96.00	96.97	98	2	4	96
PatchCraft	90.50	84.62	99.00	91.24	82	18	1	99
FreqNet	84.50	96.00	72.00	82.29	97	3	28	72

TABLE V
AI-IMAGE DETECTOR PERFORMANCE ON THE CNNDetection BENCHMARK.

Detector	Accuracy	Precision	Recall	F1-score
SAFE	82.11	75.45	95.19	84.18
NPR Deepfake	79.84	72.45	96.27	82.68
FreqNet	81.20	77.31	88.33	82.45
PatchCraft	76.41	71.73	87.19	78.71

cussion, and Conclusion and Future Work sections. It was also used to support the development of scripts for data processing, model evaluation, metric calculation, multimodal fusion, and result formatting.

All GenAI-generated suggestions, text, and code were reviewed, edited, tested, and validated by the author before being included in the dissertation. The experimental decisions, interpretation of results, final written content, and overall conclusions remain the responsibility of the author.

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TABLE VI
AI-TEXT DETECTOR PERFORMANCE ON THE MARKETPLACE DESCRIPTION DATASET.

Detector	Accuracy	Precision	Recall	F1-score	TN	FP	FN	TP
RoBERTa Base	37.50	41.01	57.00	47.70	18	82	43	57
RoBERTa Large	44.00	46.25	74.00	56.92	14	86	26	74
Binoculars	65.00	60.27	88.00	71.54	42	58	12	88
FastDetectGPT	54.00	53.64	59.00	56.19	49	51	41	59

TABLE VII
MULTIMODAL PERFORMANCE FOR EXPERIMENT 1: REAL LISTINGS VERSUS FAKE IMAGES WITH HUMAN-WRITTEN DESCRIPTIONS.

Method	Accuracy	Precision	Recall	F1-score	Specificity	TN	FP	FN	TP
NPR only	99.50	99.01	100.00	99.50	99.00	99	1	0	100
SAFE only	97.00	97.96	96.00	96.97	98.00	98	2	4	96
Binoculars (Text only)	62.00	57.79	89.00	70.08	35.00	35	65	11	89
Multimodal Majority Vote	98.00	97.06	99.00	98.02	97.00	97	3	1	99

TABLE VIII
MULTIMODAL PERFORMANCE FOR EXPERIMENT 2: REAL LISTINGS VERSUS FAKE IMAGES WITH AI-GENERATED DESCRIPTIONS.

Method	Accuracy	Precision	Recall	F1-score	Specificity	TN	FP	FN	TP
NPR only	99.50	99.01	100.00	99.50	99.00	99	1	0	100
SAFE only	97.00	97.96	96.00	96.97	98.00	98	2	4	96
Binoculars (Text only)	63.50	58.60	92.00	71.60	35.00	35	65	8	92
Multimodal Majority Vote	98.50	97.09	100.00	98.52	97.00	97	3	0	100

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